
Primitive Shape Recognition from 3D Point Clouds Using PointNet-Based Deep Learning

MSc Robotics / Advanced project

College of Engineering and Physical Sciences | School of Engineering

Presented by - Kunal Khose (3055813)

Supervisor: Dr Marco Castellani



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PROJECT AIM & OBJECTIVES

□ Aim

To train and evaluate a PointNet-based model for recognising primitive shapes from 3D point-cloud data.

□ Objectives

- Review PointNet, PointNet++ and point-cloud classification literature.
- Set up a PyTorch-based PointNet/PointNet++ model using Google colab.
- Train and test using the available APS dataset.
- Evaluate accuracy, loss and confusion matrix.
- Test robustness using APS clean & error variants.
- Explore real point-cloud testing if suitable data available and time permits.



PROBLEM STATEMENT

Why primitive shape recognition matters for robotics

- Robots need shape understanding for grasping, sorting and disassembly.
- Point clouds represent 3D object geometry using unordered x, y, z points.
- Real scans can include noise, missing points, partial views and size variation.

Research question -

How reliably can a PointNet-based model recognise primitive shapes from point-cloud data?

Focus: box, cylinder and sphere classes using APS point clouds.



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CURRENT STATE-OF-THE-ART

Method / Concept	Relevance to This Project
Deep Learning	Learns features directly from data and reduces the need for hand-designed feature extraction [1].
Point-Cloud Learning	Raw point clouds are unordered & irregular, so conventional image CNNs are not directly suitable [2].
PointNet	Processes unordered 3D point sets directly for classification and segmentation [3].
PointNet++	Extends PointNet by learning local neighbourhood features in a hierarchical way [4].

Table 1: Summary of Relevant Methods for Point-Cloud Shape Recognition

- Primitive shape recognition is useful for robotic manipulation and disassembly, but generalisation from artificial point clouds to noisy or real object scans remains a key challenge [5].



APPROACH & METHODOLOGY

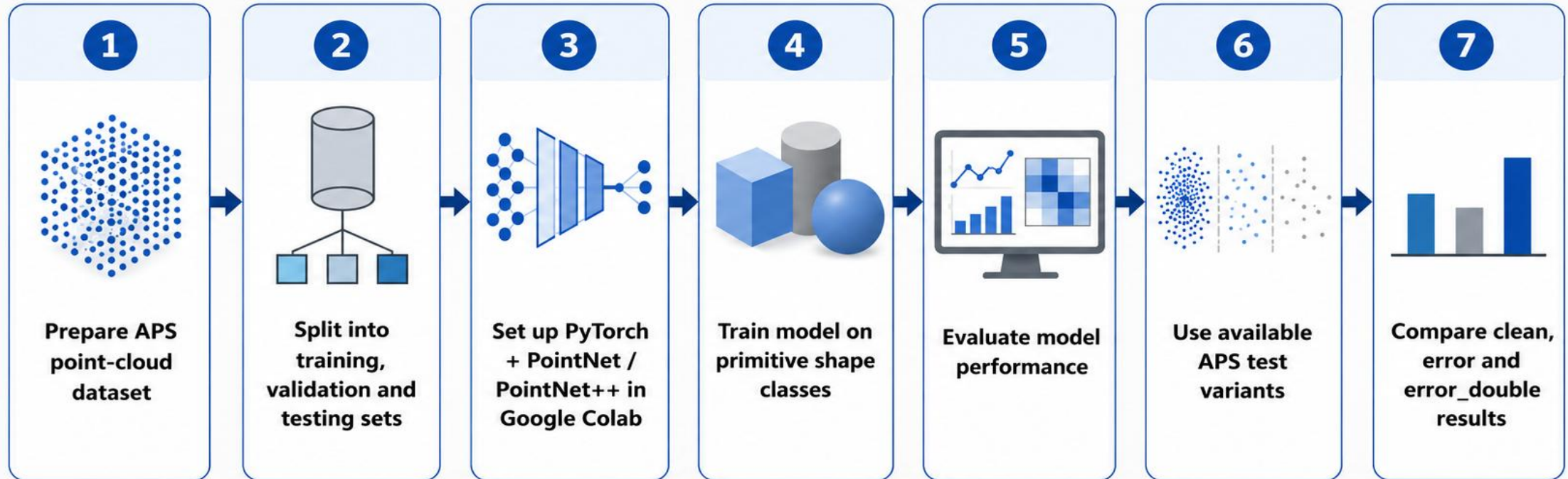


Figure 1: PointNet-Based Primitive Shape Recognition Workflow



IMPLEMENTATION

Step 1: Programming Platform

Use Python and PyTorch for implementation, with training mainly performed in Google Colab using GPU resources [3,4].

Step 2: Local Device Role

Use the Windows laptop for dataset preparation, code editing, result analysis and CPU-based inference checks.

Step 3: Model Strategy

Attempt PointNet++ as the main method, with basic PointNet as a backup or baseline if required [3,4].

Step 4: Main Coding Task

Adapt or create an APS dataloader for training, validation & testing using box, cylinder & sphere point-cloud data [5,8].

 **Main classes:** Box, cylinder and sphere [5,8].

★ **Metrics:** Accuracy, training loss, validation loss, confusion matrix and class-wise performance [2].



TECHNICAL MILESTONES

Milestone (M)	Purpose
Literature review	Understand deep learning, PointNet++ and point-cloud classification methods [1–4].
APS dataset preparation	Organise the APS point-cloud data for box, cylinder and sphere classes [8].
Model setup	Set up the PointNet/PointNet++ implementation in PyTorch using Google Colab
Dataset loader adaptation	Modify or create a dataloader so APS point-cloud files can be used for model training and testing [8].
Baseline experiment	Train and test the model on clean APS data to obtain initial classification results [3,5].
Robustness testing	Test model performance on noisy variants to assess generalisation [2,5].
Result analysis	Compare accuracy, loss curves, confusion matrix and class-wise performance [2].

Table 2: Technical Milestones and Purpose of the PointNet-Based Primitive Shape Recognition Project



EXPECTED OUTPUTS

- Prepared APS dataset structure for box, cylinder and sphere point-cloud classes [8].
- Google Colab notebook / Python scripts for PyTorch training pipeline for PointNet / PointNet++ [3,4].
- Trained PointNet-based model for primitive shape classification [3–5].
- Baseline results on the clean APS dataset [5,8].
- Robustness test results using APS error variants or modified point-cloud test sets [2,5].
- Evaluation outputs: accuracy, training/validation loss plots, confusion matrix and class-wise performance [2].
- Comparison table showing clean vs noisy / reduced-density results [2,5].
- Technical discussion of strengths, limitations and possible future improvements.



PROJECT IMPACT

Impact:

- Supporting robotic object recognition [5].
- Useful for grasping, sorting, disassembly and remanufacturing [5].
- Helps assess whether artificial point-cloud training can support real applications [5].

Limitations:

- Software-only project.
- No physical robot testing.
- Training depends on Colab/GPU availability and real-data testing will depend on dataset preparation time.

Future scope:

- Test on larger real datasets [2,7].
- Add more primitive and complex shapes [8].
- Connect recognition output to robotic grasp planning [5,7].



TECHNICAL GANTT CHART

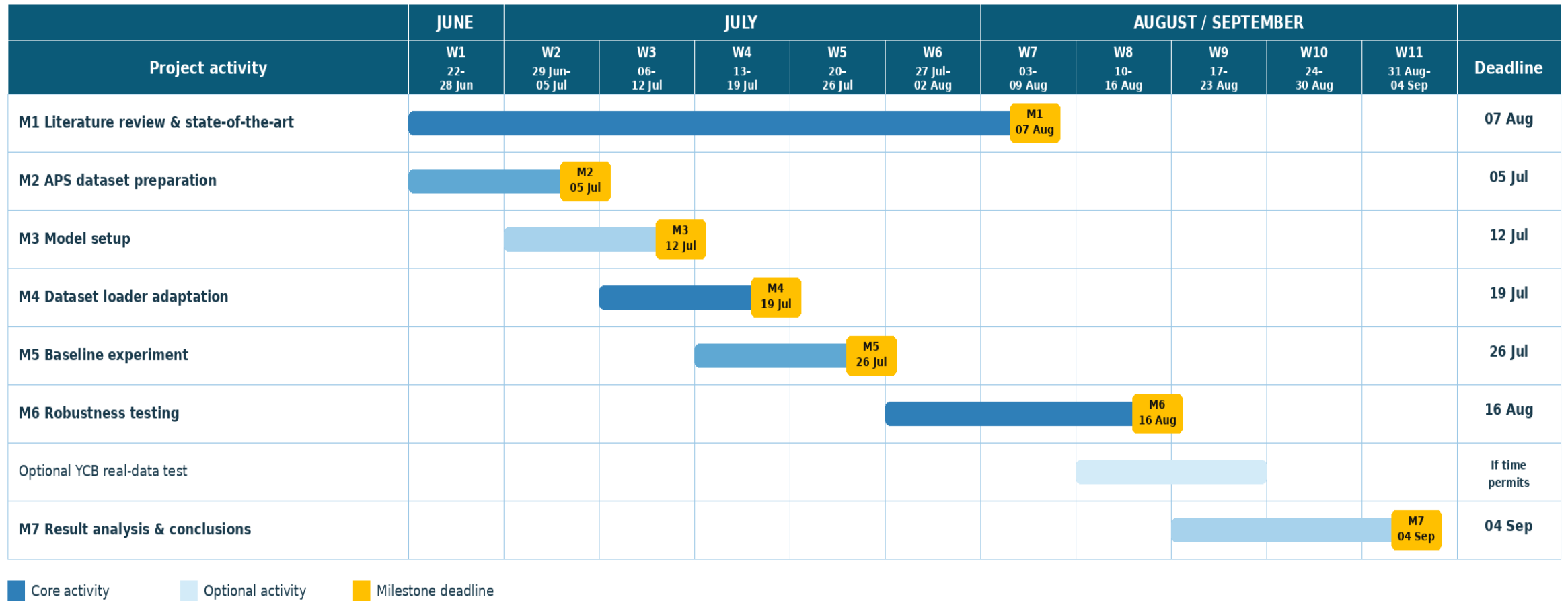


Figure 2: Technical Gantt Chart for the Project Timeline

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Thank you



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